

Blackmagic SDI Camera Control Protocol

Version 1.6.2

If you are a software developer you can use the Blackmagic SDI to construct devices that integrate with our products. Here at Blackmagic Design, our approach is to open up our protocols and we eagerly look forward to seeing what you come up with!

Overview

This document describes an extensible protocol for sending a unidirectional stream of small control messages embedded in the non-active picture region of a digital video stream. The video stream containing the protocol stream may be broadcast to a number of devices. Device addressing is used to allow the sender to specify which device each message is directed to.

Assumptions

Alignment and padding constraints are explicitly described in the protocol document. Bit fields are packed from LSB first. Message groups, individual messages and command headers are defined as, and can be assumed to be, 32 bit aligned.

Blanking Encoding

A message group is encoded into a SMPTE 291M packet with DID/SDID x51/x53 in the active region of VANC line 16.

Message Grouping

Up to 32 messages may be concatenated and transmitted in one blanking packet up to a maximum of 255 bytes payload. Under most circumstances, this should allow all messages to be sent with a maximum of one frame latency.

If the transmitting device queues more bytes of message packets than can be sent in a single frame, it should use heuristics to determine which packets to prioritize and send immediately. Lower priority messages can be delayed to later frames, or dropped entirely as appropriate.

Abstract Message Packet Format

Every message packet consists of a three byte header followed by an optional variable length data block. The maximum packet size is 64 bytes.

Destination device (uint8)	Device addresses are represented as an 8 bit unsigned integer. Individual devices are numbered 0 through 254 with the value 255 reserved to indicate a broadcast message to all devices.
Command length (uint8)	The command length is an 8 bit unsigned integer which specifies the length of the included command data. The length does NOT include the length of the header or any trailing padding bytes.
Command id (uint8)	The command id is an 8 bit unsigned integer which indicates the message type being sent. Receiving devices should ignore any commands that they do not understand. Commands 0 through 127 are reserved for commands that apply to multiple types of devices. Commands 128 through 255 are device specific.
Reserved (uint8)	This byte is reserved for alignment and expansion purposes. It should be set to zero.

Command data (uint8[])	The command data may contain between 0 and 60 bytes of data. The format of the data section is defined by the command itself.
Padding (uint8[])	Messages must be padded up to a 32 bit boundary with 0x0 bytes. Any padding bytes are NOT included in the command length.

Receiving devices should use the destination device address and or the command identifier to determine which messages to process. The receiver should use the command length to skip irrelevant or unknown commands and should be careful to skip the implicit padding as well.

Defined Commands

Command 0 : change configuration

Category (uint8)	The category number specifies one of up to 256 configuration categories available on the device.
Parameter (uint8)	The parameter number specifies one of 256 potential configuration parameters available on the device. Parameters 0 through 127 are device specific parameters. Parameters 128 through 255 are reserved for parameters that apply to multiple types of devices.
Data type (uint8)	The data type specifies the type of the remaining data. The packet length is used to determine the number of elements in the message. Each message must contain an integral number of data elements.

Currently defined values are:

0: void/boolean	A void value is represented as a boolean array of length zero. The data field is a 8 bit value with 0 meaning false and all other values meaning true.
1: signed byte	Data elements are signed bytes
2: signed 16 bit integer	Data elements are signed 16 bit values
3: signed 32 bit integer	Data elements are signed 32 bit values
4: signed 64 bit integer	Data elements are signed 64 bit values
5: UTF-8 string	Data elements represent a UTF-8 string with no terminating character.

Data types 6 through 127 are reserved.

128: signed 5.11 fixed point	Data elements are signed 16 bit integers representing a real number with 5 bits for the integer component and 11 bits for the fractional component. The fixed point representation is equal to the real value multiplied by 2^{11} . The representable range is from -16.0 to 15.9995 ($15 + 2047/2048$).
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Data types 129 through 255 are available for device specific purposes.

Operation type (uint8)	The operation type specifies what action to perform on the specified parameter. Currently defined values are:
0: assign value	The supplied values are assigned to the specified parameter. Each element will be clamped according to its valid range. A void parameter may only be 'assigned' an empty list of boolean type. This operation will trigger the action associated with that parameter. A boolean value may be assigned the value zero for false, and any other value for true.
1: offset/toggle value	Each value specifies signed offsets of the same type to be added to the current parameter values. The resulting parameter value will be clamped according to their valid range. It is not valid to apply an offset to a void value. Applying any offset other than zero to a boolean value will invert that value.
Operation types 2 through 127 are reserved.	

Operation types 128 through 255 are available for device specific purposes.

Data (void)	The data field is 0 or more bytes as determined by the data type and number of elements.
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The category, parameter, data type and operation type partition a 24 bit operation space.

Group	ID	Parameter	Type	Index	Minimum	Maximum	Interpretation
Lens	0.0	Focus	fixed16	–	0.0	1.0	0.0 = near, 1.0 = far
	0.1	Instantaneous autofocus	void	–	–	–	trigger instantaneous autofocus
	0.2	Aperture (f-stop)	fixed16	–	-1.0	16.0	Aperture Value (where fnumber = $\sqrt{2^{AV}}$)
	0.3	Aperture (normalised)	fixed16	–	0.0	1.0	0.0 = smallest, 1.0 = largest
	0.4	Aperture (ordinal)	int16	–	0	n	Steps through available aperture values from minimum (0) to maximum (n)
	0.5	Instantaneous auto aperture	void	–	–	–	trigger instantaneous auto aperture
	0.6	Optical image stabilisation	boolean	–	–	–	true = enabled, false = disabled
	0.7	Set absolute zoom (mm)	int16	–	0	max	Move to specified focal length in mm, from minimum (0) to maximum (max)
	0.8	Set absolute zoom (normalised)	fixed16	–	0.0	1.0	Move to specified focal length: 0.0 = wide, 1.0 = tele
	0.9	Set continuous zoom (speed)	fixed16	–	-1.0	+1.0	Start/stop zooming at specified rate: -1.0 = zoom wider fast, 0.0 = stop, +1 = zoom tele fast

Group	ID	Parameter	Type	Index	Minimum	Maximum	Interpretation
Video	1.0	Video mode	int8	[0] = frame rate	–	–	fps as integer (eg 24, 25, 30, 50, 60)
				[1] = M-rate	–	–	0 = regular, 1 = M-rate
				[2] = dimensions	–	–	0 = NTSC, 1 = PAL, 2 = 720, 3 = 1080, 4 = 2kDCI, 5 = 2k16:9, 6 = UHD, 7 = 3k Anamorphic, 8 = 4k DCI, 9 = 4k 16:9, 10 = 4.6k 2.4:1, 11 = 4.6k
				[3] = interlaced	–	–	0 = progressive, 1 = interlaced
				[4] = Color space	–	–	0 = YUV
	1.1	Gain (up to Camera 4.9)	int8		1	128	1x, 2x, 4x, 8x, 16x, 32x, 64x, 128x gain
	1.2	Manual White Balance	int16	[0] = color temp	2500	10000	Color temperature in K
			int16	[1] = tint	-50	50	tint
	1.3	Set auto WB	void	–	–	–	Calculate and set auto white balance
	1.4	Restore auto WB	void	–	–	–	Use latest auto white balance setting
	1.5	Exposure (us)	int32		1	42000	time in us
	1.6	Exposure (ordinal)	int16	–	0	n	Steps through available exposure values from minimum (0) to maximum (n)
	1.7	Dynamic Range Mode	int8 enum	–	0	2	0 = film, 1 = video, 2 = extended video
	1.8	Video sharpening level	int8 enum	–	0	3	0 = off, 1 = low, 2 = medium, 3 = high
	1.9	Recording format	int16	[0] = file frame rate	–	–	fps as integer (eg 24, 25, 30, 50, 60, 120)
				[1] = sensor frame rate	–	–	fps as integer, valid when sensor-off-speed set (eg 24, 25, 30, 33, 48, 50, 60, 120), no change will be performed if this value is set to 0
				[2] = frame width	–	–	in pixels
				[3] = frame height	–	–	in pixels
				[4] = flags	–	–	[0] = file-M-rate
					–	–	[1] = sensor-M-rate, valid when sensor-off-speed-set
					–	–	[2] = sensor-off-speed
					–	–	[3] = interlaced
					–	–	[4] = windowed mode
	1.10	Set auto exposure mode	int8	–	0	4	0 = Manual Trigger, 1 = Iris, 2 = Shutter, 3 = Iris + Shutter, 4 = Shutter + Iris
	1.11	Shutter angle	int32	–	100	36000	Shutter angle in degrees, multiplied by 100
	1.12	Shutter speed	int32	–	Current sensor frame rate	5000	Shutter speed value as a fraction of 1, so 50 for 1/50th of a second
	1.13	Gain	int8	–	-128	127	Gain in decibel (dB)
	1.14	ISO	int32	–	0	2147483647	ISO value
	1.15	Display LUT	int8	[0] = selected LUT	–	–	0 = None, 1 = Custom, 2 = film to video, 3 = film to extended video
				[1] = enabled or not	–	–	0 = Not enabled, 1 = Enabled

Group	ID	Parameter	Type	Index	Minimum	Maximum	Interpretation
	1.16	ND Filter Stop	fixed16	[0] = stop	0.0	15.0	filter power, as f-stop
				[1] = display mode	–	–	0 = stop 1 = density 2 = transmittance
Audio	2.0	Mic level	fixed16	–	0.0	1.0	0.0 = minimum, 1.0 = maximum
	2.1	Headphone level	fixed16	–	0.1	1.0	0.0 = minimum, 1.0 = maximum
	2.2	Headphone program mix	fixed16	–	0.1	1.0	0.0 = minimum, 1.0 = maximum
	2.3	Speaker level	fixed16	–	0.1	1.0	0.0 = minimum, 1.0 = maximum
	2.4	Input type	int8	–	0	3	0 = internal mic, 1 = line level input, 2 = low mic level input, 3 = high mic level input
	2.5	Input levels	fixed16	[0] ch0	0.0	1.0	0.0 = minimum, 1.0 = maximum
				[1] ch1	0.0	1.0	0.0 = minimum, 1.0 = maximum
	2.6	Phantom power	boolean	–	–	–	true = powered, false = not powered
Output	3.0	Overlay enables	uint16 bit field	[0] = bit field	–	–	bit flags: [0] = display status, [1] = display frame guides [2] = clean feed Some cameras don't allow separate control of frame guides and status overlays.
			uint16 bit field	[1] = target displays bit field	–	–	bit flags: [0] = LCD [1] = HDMI [2] = EVF [3] = Main SDI [4] = Front SDI
	3.1	Frame guides style (Camera 3.x)	int8	–	0	8	0 = HDTV, 1 = 4:3, 2 = 2.4:1, 3 = 2.39:1, 4 = 2.35:1, 5 = 1.85:1, 6 = thirds
	3.2	Frame guides opacity (Camera 3.x)	fixed16	–	0.1	1.0	0.0 = transparent, 1.0 = opaque
	3.3	Overlays (replaces .1 and .2 above from Cameras 4.0)	int8	[0] = frame guides style	–	–	0 = off, 1 = 2.4:1, 2 = 2.39:1, 3 = 2.35:1, 4 = 1.85:1, 5 = 16:9, 6 = 14:9, 7 = 4:3, 8 = 2:1, 9 = 4:5, 10 = 1:1
				[1] = frame guide opacity	0	100	0 = transparent, 100 = opaque
				[2] = safe area percentage	0	100	percentage of full frame used by safe area guide (0 means off)
				[3] = grid style	–	–	bit flags: [0] = display thirds, [1] = display cross hairs, [2] = display center dot, [3] = display horizon

Group	ID	Parameter	Type	Index	Minimum	Maximum	Interpretation
Display	4.0	Brightness	fixed16	–	0.0	1.0	0.0 = minimum, 1.0 = maximum
	4.1	Exposure and focus tools	uint16 bit field	[0] = bit field	–	–	bit flags: [0] = Zebra [1] = Focus Assist [2] = False Color
			uint16 bit field	[1] = target displays bit field	–	–	bit flags: [0] = LCD [1] = HDMI [2] = EVF [3] = Main SDI [4] = Front SDI
	4.2	Zebra level	fixed16	–	0.0	1.0	0.0 = minimum, 1.0 = maximum
	4.3	Peaking level	fixed16	–	0.0	1.0	0.0 = minimum, 1.0 = maximum
	4.4	Color bar enable	int8	–	0	30	0 = disable bars, 1-30 = enable bars with timeout (seconds)
	4.5	Focus Assist	int8	[0] = focus assist method	–	–	0 = Peak, 1 = Colored lines
				[1] = focus line color	–	–	0 = Red, 1 = Green, 2 = Blue, 3 = White, 4 = Black
	4.6	Program return feed enable	int8	–	0	30	0 = disable, 1-30 = enable with timeout (seconds)
	4.7	Timecode Source	signed byte	[0] = source	–	–	0 = Clip, 1 = Timecode
Tally	5.0	Tally brightness	fixed16	–	0.0	1.0	Sets the tally front and tally rear brightness to the same level. 0.0 = minimum, 1.0 = maximum
	5.1	Front tally brightness	fixed16	–	0.0	1.0	Sets the tally front brightness. 0.0 = minimum, 1.0 = maximum
	5.2	Rear tally brightness	fixed16	–	0.0	1.0	Sets the tally rear brightness. 0.0 = minimum, 1.0 = maximum Tally rear brightness cannot be turned off
Reference	6.0	Source	int8 enum	–	0	2	0 = internal, 1 = program, 2 = external
	6.1	Offset	int32	–	–	–	+/- offset in pixels

Group	ID	Parameter	Type	Index	Minimum	Maximum	Interpretation
Confi- guration	7.0	Real Time Clock	int32	[0] time	—	—	BCD - HHMMSSFF (UCT)
				[1] date	—	—	BCD - YYYYMMDD
	7.1	System language	string	[0-1]	—	—	ISO-639-1 two character language code
	7.2	Timezone	int32	—	—	—	Minutes offset from UTC
	7.3	Location	int64	[0] latitude	—	—	BCD - s0DDdddddddddd where s is the sign: 0 = north (+), 1 = south (-); DD degrees, ddddddddddd decimal degrees
				[1] longitude	—	—	BCD - sDDDDdddddddddd where s is the sign: 0 = west (-), 1 = east (+); DDD degrees, ddddddddddd decimal degrees
Color Correction	8.0	Lift Adjust	fixed16	[0] red	-2.0	2.0	default 0.0
				[1] green	-2.0	2.0	default 0.0
				[2] blue	-2.0	2.0	default 0.0
				[3] luma	-2.0	2.0	default 0.0
	8.1	Gamma Adjust	fixed16	[0] red	-4.0	4.0	default 0.0
				[1] green	-4.0	4.0	default 0.0
				[2] blue	-4.0	4.0	default 0.0
				[3] luma	-4.0	4.0	default 0.0
	8.2	Gain Adjust	fixed16	[0] red	0.0	16.0	default 1.0
				[1] green	0.0	16.0	default 1.0
				[2] blue	0.0	16.0	default 1.0
				[3] luma	0.0	16.0	default 1.0
	8.3	Offset Adjust	fixed16	[0] red	-8.0	8.0	default 0.0
				[1] green	-8.0	8.0	default 0.0
				[2] blue	-8.0	8.0	default 0.0
				[3] luma	-8.0	8.0	default 0.0
	8.4	Contrast Adjust	fixed16	[0] pivot	0.0	1.0	default 0.5
				[1] adj	0.0	2.0	default 1.0
	8.5	Luma mix	fixed16	—	0.0	1.0	default 1.0
	8.6	Color Adjust	fixed16	[0] hue	-1.0	1.0	default 0.0
				[1] sat	0.0	2.0	default 1.0
	8.7	Correction Reset Default	void	—	—	—	reset to defaults

Group	ID	Parameter	Type	Index	Minimum	Maximum	Interpretation
Media	10.0	Codec	int8 enum	[0] = basic codec	–	–	0 = CinemaDNG, 1 = DNxHD, 2 = ProRes, 3 = Blackmagic RAW
				[1] = code variant	–	–	CinemaDNG: 0 = uncompressed, 1 = lossy 3:1, 2 = lossy 4:1
					–	–	ProRes: 0 = HQ, 1 = 422, 2 = LT, 3 = Proxy, 4 = 444, 5 = 444XQ
					–	–	Blackmagic RAW: 0 = Q0, 7 = Q1, 8 = Q3, 1 = Q5, 2 = 3:1, 3 = 5:1, 4 = 8:1, 5 = 12:1
	10.1	Transport mode	int8	[0] = mode	–	–	0 = Preview, 1 = Play, 2 = Record
				[1] = speed	–	–	-ve = multiple speeds backwards, 0 = pause, +ve = multiple speeds forwards
				[2] = flags	–	–	1<<0 = loop, 1<<1 = play all, 1<<5 = disk1 active, 1<<6 = disk2 active, 1<<7 = time-lapse recording
				[3] = slot 1 storage medium	–	–	0 = CFast card, 1 = SD, 2 = SSD Recorder
				[4] = slot 2 storage medium	–	–	0 = CFast card, 1 = SD, 2 = SSD Recorder
	10.2	Playback Control	int8 enum	[0] = clip	–	–	0 = Previous, 1 = Next
	10.5	Stream	bool	[0] = enabled	–	–	true = enabled, false = disabled
	10.6	Stream Information	void bool	[0] = enabled	–	–	true = enabled, false = disabled
	10.7	Stream Display 3D LUT	void bool	[0] = enabled	–	–	true = enabled, false = disabled

Group	ID	Parameter	Type	Index	Minimum	Maximum	Interpretation
PTZ Control	11.0	Pan/Tilt Velocity	fixed 16	[0] = pan velocity	-1.0	1.0	-1.0 = full speed left, 1.0 = full speed right
				[1] = tilt velocity	-1.0	1.0	-1.0 = full speed down, 1.0 = full speed up
	11.1	Memory Preset	int8 enum	[0] = preset command	–	–	0 = reset, 1 = store location, 2 = recall location
			int8	[1] = preset slot	0	5	–

Example Protocol Packets

Operation	Packet Length	Byte															
		0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
		header				command				data							
		destination	length	command	reserved	category	parameter	type	operation								
trigger instantaneous auto focus on camera 4	8	4	4	0	0	0	1	0	0								
turn on OIS on all cameras	12	255	5	0	0	0	6	0	0	1	0	0	0				
set exposure to 10 ms on camera 4 (10 ms = 10000 us = 0x00002710)	12	4	8	0	0	1	5	3	0	0x10	0x27	0x00	0x00				
add 15% to zebra level (15 % = 0.15 f = 0x0133 fp)	12	4	6	0	0	4	2	128	1	0x33	0x01	0	0				
select 1080p 23.98 mode on all cameras	16	255	9	0	0	1	0	1	0	24	1	3	0	0	0	0	0
subtract 0.3 from gamma adjust for green & blue (-0.3 ~ = 0xfd9a fp)	16	4	12	0	0	8	1	128	1	0	0	0x9a	0xfd	0x9a	0xfd	0	0
all operations combined	76	4	4	0	0	0	1	0	0	255	5	0	0	0	6	0	0
		1	0	0	0	4	8	0	0	1	5	3	0	0x10	0x27	0x00	0x00
		4	6	0	0	4	2	128	1	0x33	0x01	0	0	255	9	0	0
		1	0	1	0	24	1	3	0	0	0	0	0	4	12	0	0
		8	1	128	1	0	0	0x9a	0xfd	0x9a	0xfd	0	0				

Blackmagic Embedded Tally Control Protocol

Version 1.0 (30/04/14)

This section is for third party developers or anybody who may wish to add support for the Blackmagic Embedded Tally Control Protocol to their products or system. It describes the protocol for sending tally information embedded in the non-active picture region of a digital video stream.

Data Flow

A master device such as a broadcast switcher embeds tally information into its program feed which is broadcast to a number of slave devices such as cameras or camera controllers. The output from the slave devices is typically fed back to the master device, but may also be sent to a video monitor.

The primary flow of tally information is from the master device to the slaves. Each slave device may use its device id to extract and display the relevant tally information.

Slave devices pass through the tally packet on their output and update the monitor tally status, so that monitor devices connected to that individual output may display tally status without knowledge of the device id they are monitoring.

Assumptions

Any data alignment / padding is explicit in the protocol. Bit fields are packed from LSB first.

Blanking Encoding

One tally control packet may be sent per video frame. Packets are encoded as a SMPTE 291M packet with DID/SDID x51/x52 in the active region of VANC line 15. A tally control packet may contain up to 256 bytes of tally information.

Packet Format

Each tally status consists of 4 bits of information:

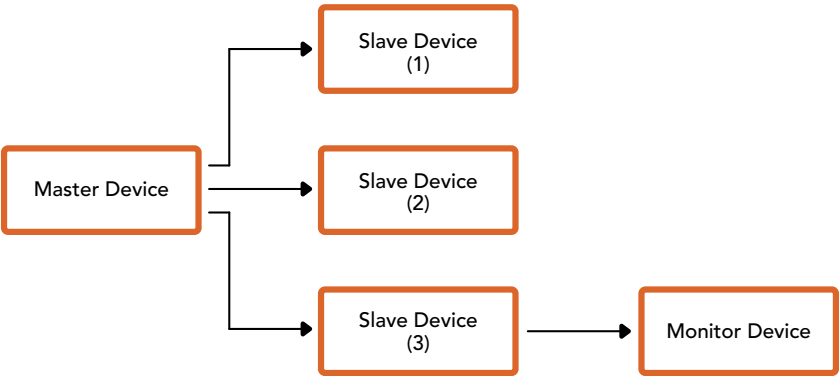
```
uint4
bit 0:    program tally status (0=off, 1=on)
bit 1:    preview tally status (0=off, 1=on)
bit 2-3:  reserved (0x0)
```

The first byte of the tally packet contains the monitor device tally status and a version number.

Subsequent bytes of the tally packet contain tally status for pairs of slave devices. The master device sends tally status for the number of devices configured/supported, up to a maximum of 510.

```
struct tally
uint8
bit 0:    monitor device program tally status (0=off, 1=on)
bit 1:    monitor device preview tally status (0=off, 1=on)
bit 2-3:  reserved (0b00)
bit 4-7:  protocol version (0b0000)
uint8[0]
bit 0:    slave device 1 program tally status (0=off, 1=on)
bit 1:    slave device 1 device preview tally status (0=off, 1=on)
bit 2-3:  reserved (0b00)
bit 4:    slave device 2 program tally status (0=off, 1=on)
bit 5:    slave device 2 preview tally status (0=off, 1=on)
bit 6-7:  reserved (0b00)
```

uint8[1]
 bit 0: slave device 3 program tally status (0=off, 1=on)
 bit 1: slave device 3 device preview tally status (0=off, 1=on)
 bit 2-3: reserved (0b00)
 bit 4: slave device 4 program tally status (0=off, 1=on)
 bit 5: slave device 4 preview tally status (0=off, 1=on)
 bit 6-7: reserved (0b00)
 ...



Byte	7 MSB	6	5	4	3	2	1	0 LSB
0	Version (0b0)	Version (0b0)	Version (0b0)	Version (0b0)	Reserved (0b0)	Reserved (0b0)	Monitor Preview	Monitor Program
1	Reserved (0b0)	Reserved (0b0)	Slave 1 Preview	Slave 1 Program	Reserved (0b0)	Reserved (0b0)	Slave 0 Preview	Slave 0 Program
2	Reserved (0b0)	Reserved (0b0)	Slave 3 Preview	Slave 3 Program	Reserved (0b0)	Reserved (0b0)	Slave 2 Preview	Slave 2 Program
3	...							

Visca Commands for PTZ control via SDI

Pan-tiltDrive	Up	8x 01 06 01 VV WW 03 01 FF	VV: Pan speed 01 to 18 WW: Tilt speed 01 to 17 YYYY: Pan position F725 to 08DB (center 0000) ZZZZ: Tilt position FE70 to 04B0 (image flip: OFF) (center 0000) Tilt position FB50 to 0190 (image flip: ON) (center 0000)
	Down	8x 01 06 01 VV WW 03 02 FF	
	Left	8x 01 06 01 VV WW 01 03 FF	
	Right	8x 01 06 01 VV WW 02 03 FF	
	UpLeft	8x 01 06 01 VV WW 01 01 FF	
	UpRight	8x 01 06 01 VV WW 02 01 FF	
	DownLeft	8x 01 06 01 VV WW 01 02 FF	
	DownRight	8x 01 06 01 VV WW 02 02 FF	
	Stop	8x 01 06 01 VV WW 03 03 FF	
	AbsolutePosition	8x 01 06 02 VV WW 0Y 0Y 0Y 0Y 0Z 0Z 0Z 0Z FF	
	RelativePosition	8x 01 06 03 VV WW 0Y 0Y 0Y 0Y 0Z 0Z 0Z 0Z FF	
	Home	0Y 0Y 0Y 0Y 0Z 0Z 0Z 0Z FF	
	Reset	8x 01 06 05 FF	
CAM_Memory	Reset	8x 01 04 3F 00 0p FF	p: Memory number (=0 to 5) Corresponds to 1 to 6 on the remote commander.
	Set	8x 01 04 3F 01 0p FF	
	Recall	8x 01 04 3F 02 0p FF	

Compatible motorized heads include the following:

- KXWell KT-PH180BMD
- PTZOptics PT-Broadcaster
 - RUSHWORKS PTX Model 1

Blackmagic Bluetooth Camera Control

Blackmagic cameras with Bluetooth LE implement a variety of features and commands that allow users to control their cameras wirelessly. Developers have full access to these features for their custom applications.

The following services and characteristics describe the full range of communication options that are available to the developer.

Service: Device Information Service

UUID: 180A

Characteristics

Camera Manufacturer

UUID: 2A29

Read the name of the manufacturer (always “Blackmagic Design”).

Camera Model

UUID: 2A24

Read the name of the camera model (eg. “URSA Mini Pro”).

Service: Blackmagic Camera Service

UUID: 291D567A-6D75-11E6-8B77-86F30CA893D3

Characteristics

Outgoing Camera Control (encrypted)

UUID: 5DD3465F-1AEE-4299-8493-D2ECA2F8E1BB

Send Camera Control messages.

These messages are identical to those described in the Blackmagic SDI Camera Control Protocol section above. Please read that section for a list of supported messages and required formatting information.

For an example of how packets are structured, please see the ‘example protocol packets’ section in this document.

Incoming Camera Control (encrypted)

UUID: B864E140-76A0-416A-BF30-5876504537D9

Request notifications for this characteristic to receive Camera Control messages from the camera.

These messages are identical to those described in the Blackmagic SDI Camera Control Protocol section above. Please read that section for a list of supported messages and required formatting information.

Timecode (encrypted)

UUID: 6D8F2110-86F1-41BF-9AFB-451D87E976C8

Request notifications for this characteristic to receive timecode updates.

Timecode (HH:MM:SS:mm) is represented by a 32-bit BCD number: (eg. 09:12:53:10 = 0x09125310)

Camera Status (encrypted)

UUID: 7FE8691D-95DC-4FC5-8ABD-CA74339B51B9

Request notifications for this characteristic to receive camera status updates.

The camera status is represented by flags contained in an 8-bit integer:

None	= 0x00
Camera Power On	= 0x01
Connected	= 0x02
Paired	= 0x04
Versions Verified	= 0x08
Initial Payload Received	= 0x10
Camera Ready	= 0x20

Send a value of 0x00 to power a connected camera off.

Send a value of 0x01 to power a connected camera on.

Device Name

UUID: FFAC0C52-C9FB-41A0-B063-CC76282EB89C

Send a device name to the camera (max. 32 characters).

The camera will display this name in the Bluetooth Setup Menu.

Protocol Version

UUID: 8F1FD018-B508-456F-8F82-3D392BEE2706

Read this value to determine the camera's supported CCU protocol version.

NOTE Encrypted characteristics can only be used once a device has successfully bonded or paired with the Blackmagic Camera. Once a connection has been established, any attempt to write to an encrypted characteristic will initiate bonding. For example, writing a 'Camera Power On' (0x01) message to the Camera Status characteristic.

Once bonding is initiated, the camera will display a 6-digit pin in the Bluetooth Setup Menu. Enter this pin on your device to establish an encrypted connection. The device will now be able to read, write and receive notifications from encrypted characteristics.